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Week - 10
Adversarial search
Implement Alpha - Beta Pruning.

Algorithm :

1. Start at the root node (current game start)
   The current Player is either max or min
2. Intialize
   - $\alpha = -\infty$
   - $\beta = +\infty$
3. If terminal node (end of game):
   → Return the utility (score) of that node
4. If it's a max player:
   - Set value = $-\infty$
   - For each child of this node:
     1) Compute child_value = Alphabeta (child, depth -1, α, β, False)
     2) update value = max (value, child_value)
     3) update α = max (value, child_value)
     4) if $\alpha \geq \beta$, then break → (Prune remaining branches).
   - Return value.
5. If its a min player:
   - Set value = $+\infty$
   - For each ~~child_value~~ =
     1) Alphabeta (child, depth -1, α, β, True)
     2) update value = min (value, child_value)
     3) update β = min (β, value)
     4) if $\alpha \geq \beta$, then break → (Prune remaining branches)
   - Return value.

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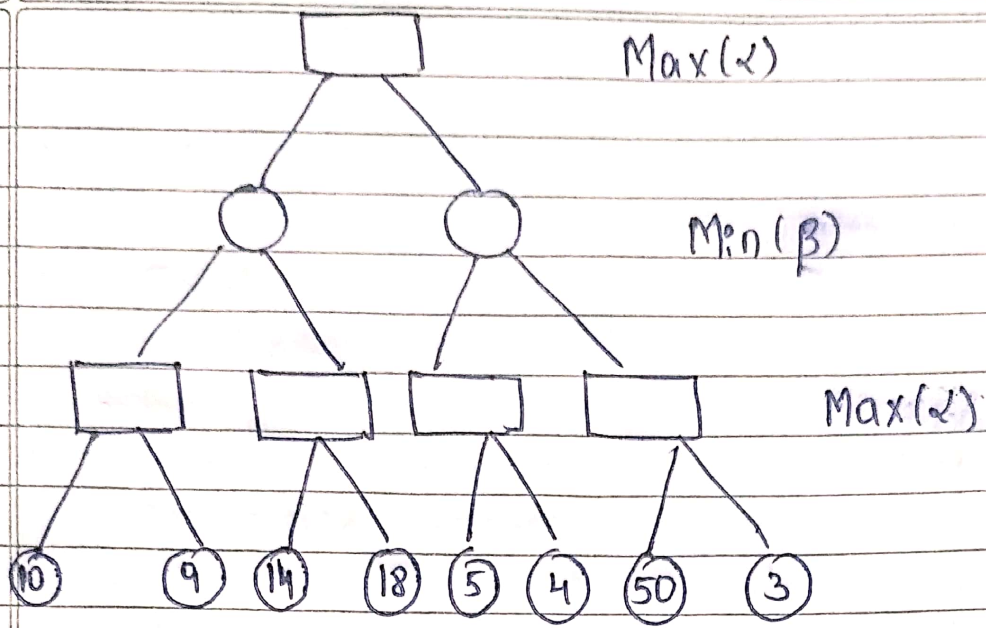
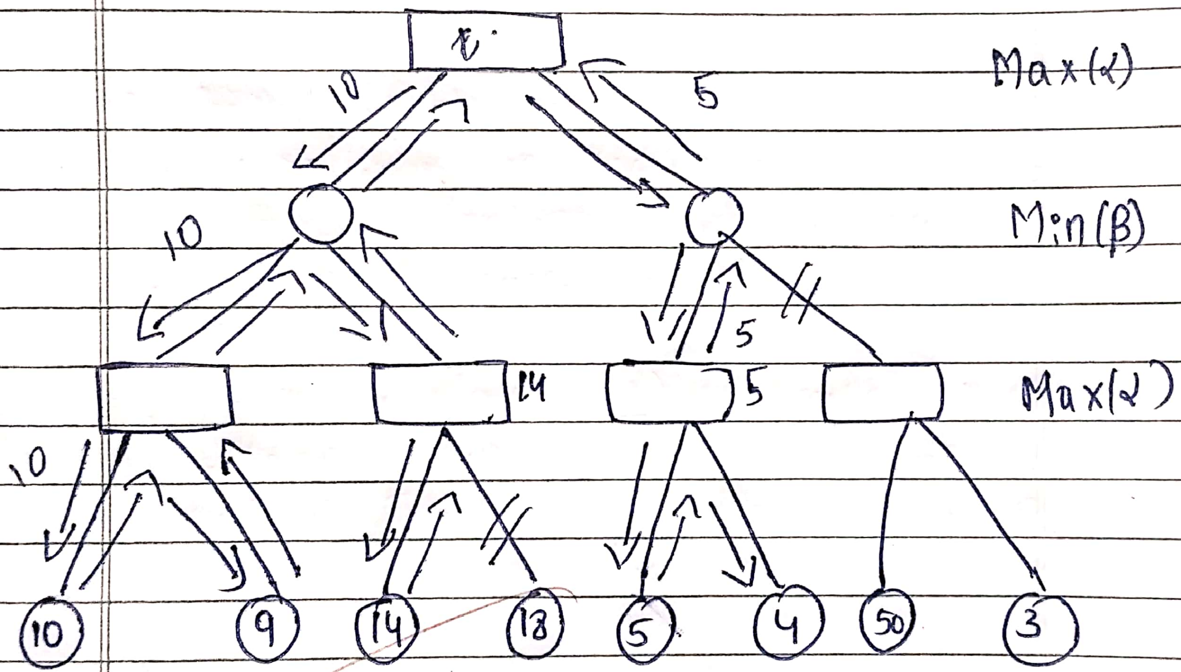


Solution:



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